Junyan (Ryan) Zhou

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Research Dr, Durham NC
https://junyanzhou.github.io/

BACKGROUND

Duke University Aug '25 - Present
Master of Science in Mechanical Engineering(Robotics) GPA: 4.0/4.0

General Robotics Lab at Duke University May '26 - Aug '26
Research Assistant

Wuhan University of Technology Sep '21 - Jun '25
Bachelor of Engineering in Automotive Engineering GPA: 3.99/5.0(5%)

RESEARCH EXPERIENCE

Geometry-Robust Tri-Axis Tactile Sensing with Improved Signal Decoupling ***May 2026 - Present***
Independent Researcher, advised by Prof. Boyuan Chen

- Developed a tactile sensor, integrating mechanical structures, custom PCBs, and embedded signal processing.
- Reproduced *eFlesh* as a baseline and conducted comparative sensor characterization using Franka Research 3.
- Evaluating sensor performance via manipulation tasks using reinforcement learning and tactile latent distillation.

COURSE PROJECT

External-Support-Aware Humanoid Control for Whole-Body Manipulation ***Aug 2026 - Present***
Group member

- Developed a contact-constrained reference-generation pipeline for Unitree G1 using Crocoddyl and Pinocchio, incorporating joint limits, collision screening, and contact-force feasibility checks.
- Adapted an existing teacher–student reinforcement learning framework for Cartesian foot-reaching commands, using proprioceptive history and learned state and joint-target estimators.

Desktop-Scale Bipedal Robot ***Jan 2026 - Apr 2026***
Individual Contributor

- Designed and developed a desktop-scale bipedal robot end-to-end, achieving the highest walking speed and longest distance in the course's final competition.

WORK EXPERIENCE

General Robotics Lab at Duke University ***May 2026 - Aug 2026***
Research Assistant

- Developed a tactile sensor, integrating mechanical structures, custom PCBs, and embedded signal processing.

VOYAH Automotive Technology Co., Ltd. ***Jan 2025 - Apr 2025***
Test Engineer(intern)

- Applied digital filtering and spectral analysis to in-cabin audio signals to evaluate noise interference and speech recognition robustness.

AWARDS & SCHOLARSHIP

- Outstanding Graduate Student, '25.
- Merit Student of the University ('21 - '24).
- University 1st-class / 3rd-class Scholarship: ('21 - '25).
- Excellent Student in Sunshine Sport Program '24.

TECHNICAL SKILLS

Mechanical Engineering: CAD(SolidWorks), CAE(ABAQUS), 3D Printing (Bambu Lab), Soldering
Electrical Engineering: PCB Design, Electrical System Integration, Embedded Systems (Sensor Acquisition, Motor Control, Serial Communication)
Computer Science: Python, C++, MATLAB, ROS2, MuJoCo, Linux

OTHERS

- **Languages**: English (C1), Mandarin (Native).